

COMPONENT LIST

Transformer-Based Dual Regulated DC Power Supply (9V & 12V)

The following component list is prepared from the submitted project report. The report specifies the main hardware used for the dual regulated power supply.

1. Components Used

S. No.	Component	Specification / Rating	Quantity	Purpose
1	Step-down Transformer	12-0-12V, 1A, center-tapped	1	Steps down AC mains voltage and provides isolation
2	Rectifier Diode	1N4007	4	Full-wave bridge rectification
3	Filter Capacitor	1000 μ F / 25V electrolytic	1	Smooths rectified DC and reduces ripple
4	Voltage Regulator IC	LM7809	1	Provides regulated 9V DC output
5	Voltage Regulator IC	LM7812	1	Provides regulated 12V DC output
6	LED Indicator	1.5V indicator LED	2	Indicates the regulated outputs
7	Resistor	1 k Ω	2	Limits current through the LED indicators
8	PCB	General-purpose PCB	1	Mounts and organizes circuit components
9	Connecting Wires	Insulated hookup wires	As required	Provides electrical connections between components

2. Main Circuit Stages

AC Input → Step-Down Transformer → Bridge Rectifier → Filter Capacitor → LM7809 / LM7812 → 9V / 12V DC Outputs → LED Indicators

3. Project Output

The project is designed to provide two regulated DC outputs: **9V DC** through LM7809 and **12V DC** through LM7812.

Source Note

Component specifications are based on the supplied project report, particularly the component list and component-description sections.